

A SEPARABLY DETERMINED IDEAL OBSTRUCTION FOR UNIVERSAL NORM ATTAINMENT

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ABSTRACT. We record a partial extension of Dantas–Jung–Martinez–Cervantes’ criterion for the existence of non-norm-attaining operators. If $\mathcal{A}$ is a norm-closed operator ideal separably determined in the domain variable, E is reflexive, and the pair (E, F) has the bounded $\mathcal{A}$ -approximation property, then universal norm-attainment for operators $E \rightarrow F$ forces $\mathcal{L}(E, F) = \mathcal{A}(E, F)$. Thus every proper such ideal yields a non-norm-attaining operator under bounded $\mathcal{A}$ -approximation. The result recovers the compact-operator case of arXiv:2102.06452 when $\mathcal{A} = \mathcal{K}$.

1. SOURCE PROBLEM

The source paper is:

Sheldon Dantas, Mingu Jung, Gonzalo Martinez-Cervantes, *On the existence of non-norm-attaining operators*, arXiv:2102.06452.

The paper quotes the following problem of Ostrovskii and the related theorem.

from Lemma 3.4. □

M.I. Ostrovskii asked in [23, §12, pg. 65] whether there exist infinite dimensional Banach spaces on which every operator attains its norm (this question is also asked in [19, Problem 8] and [14, Problem 217]). By Holub’s Theorem, if such an infinite dimensional Banach space exists, it cannot have the AP. Theorem 3.8 below should be seen as a generalization of this fact. Let us recall that given a (norm-closed) operator ideal $\mathcal{A}$ and $\lambda \geq 1$, a Banach space E is said to have the λ - $\mathcal{A}$ -approximation property (for short, λ - $\mathcal{A}$ -AP) if the identity operator Id_E belongs to $\overline{\{T \in \mathcal{A}(E, E) : \|T\| \leq \lambda\}}^{\tau_c}$. We say that E has the bounded- $\mathcal{A}$ -AP if it has the λ - $\mathcal{A}$ -AP for some $\lambda \geq 1$. This general approximation property has been studied, for instance, in [12, 20, 25, 28].

Theorem 3.8. *If there is an infinite dimensional Banach space E such that every operator on $\mathcal{L}(E)$ attains its norm, then E does not have the bounded $\mathcal{A}$ -approximation property for any ideal $\mathcal{A}$ not containing the identity on E .*

In words, Ostrovskii asks whether there are infinite-dimensional Banach spaces on which every operator attains its norm. Dantas–Jung–Martinez–Cervantes prove, among other things, that if such a space E exists, then E has no bounded $\mathcal{A}$ -approximation property for any norm-closed operator ideal $\mathcal{A}$ not containing Id_E [1, Theorem 3.8].

The result below is a pair-level version for separably determined ideals. It does not solve Ostrovskii’s problem; it enlarges the obstruction from compact operators to a class of closed ideals for pairs (E, F) .

2. DEFINITIONS

Definition 1. Let $\mathcal{A}$ be a norm-closed operator ideal. We say that $\mathcal{A}$ is separably determined in the domain variable if for every operator $T \in \mathcal{L}(X, Y)$,

$$T \in \mathcal{A}(X, Y) \iff T|_S \in \mathcal{A}(S, Y) \text{ for every separable subspace } S \subset X.$$

Definition 2. The pair (E, F) has the bounded $\mathcal{A}$ -approximation property if there is $\lambda \geq 1$ such that every $T \in \mathcal{L}(E, F)$ belongs to

$$\lambda \overline{B_{\mathcal{A}(E, F)}^{\tau_c}},$$

where τ_c is the topology of compact convergence and $B_{\mathcal{A}(E, F)} = \{S \in \mathcal{A}(E, F) : \|S\| \leq 1\}$.

3. PROOF INTUITION

Dantas–Jung–Martinez–Cervantes show that if a norm-closed convex subset of $\mathcal{L}(E, F)$ fails to be sequentially closed in the strong operator topology, then the pair (E, F) has their James property, and hence there is a non-norm-attaining operator. Thus, under universal norm-attainment, closed convex sets of operators should be strongly sequentially closed.

For the compact ideal, their paper converts SOT non-closure of the compact unit ball into SOT sequential non-closure by passing to a complemented separable subspace of the reflexive domain. The same mechanism only uses the fact that compactness is detected on separable subspaces. Therefore it works for any closed ideal $\mathcal{A}$ with the same separable-determination property. Once the ideal unit ball is SOT closed, bounded $\mathcal{A}$ -approximation leaves no room for operators outside $\mathcal{A}(E, F)$.

4. MAIN RESULT

Lemma 1. Let E be reflexive, let F be a Banach space, and let $\mathcal{A}$ be a norm-closed operator ideal separably determined in the domain variable. If $B_{\mathcal{A}(E, F)}$ is not SOT closed in $\mathcal{L}(E, F)$, then $B_{\mathcal{A}(E, F)}$ is not SOT sequentially closed.

Proof. Choose $T \in \overline{B_{\mathcal{A}(E, F)}}^{\text{SOT}} \setminus B_{\mathcal{A}(E, F)}$. Since T is a pointwise limit of contractions, $\|T\| \leq 1$. Hence $T \notin \mathcal{A}(E, F)$.

Because $\mathcal{A}$ is separably determined, there is a separable subspace $S \subset E$ such that $T|_S \notin \mathcal{A}(S, F)$. Reflexive spaces have the separable complementation property [1, Lemma 3.5 and its proof]; hence there is a complemented separable subspace $Z \subset E$ with $S \subset Z$. Let $P : E \rightarrow Z$ be a bounded projection.

Since T lies in the SOT closure of $B_{\mathcal{A}(E, F)}$, the restriction $T|_Z$ lies in the SOT closure of

$$\{R|_Z : R \in B_{\mathcal{A}(E, F)}\} \subseteq B_{\mathcal{A}(Z, F)}.$$

The space Z is separable, so we can choose a sequence $(R_n) \subset B_{\mathcal{A}(E, F)}$ such that $R_n z \rightarrow Tz$ for every $z \in Z$. Let $C = \max\{1, \|P\|\}$. Then

$$S_n := C^{-1} R_n P \in B_{\mathcal{A}(E, F)}$$

and $S_n \rightarrow C^{-1}TP$ in the SOT on $\mathcal{L}(E, F)$.

The limit $C^{-1}TP$ does not belong to $\mathcal{A}(E, F)$. Indeed, if it did, then restricting to Z would give $C^{-1}T|_Z \in \mathcal{A}(Z, F)$, and then restricting again to S would give $T|_S \in \mathcal{A}(S, F)$, a contradiction. Thus $B_{\mathcal{A}(E, F)}$ is not SOT sequentially closed. $\square$

Theorem 1. Let E be reflexive and F be a Banach space. Let $\mathcal{A}$ be a norm-closed operator ideal separably determined in the domain variable. If every operator from E into F attains its norm, then $B_{\mathcal{A}(E, F)}$ is SOT closed.

Consequently, if the pair (E, F) has the bounded $\mathcal{A}$ -approximation property and every operator $E \rightarrow F$ attains its norm, then

$$\mathcal{L}(E, F) = \mathcal{A}(E, F).$$

Equivalently, if $\mathcal{A}(E, F) \neq \mathcal{L}(E, F)$ and (E, F) has the bounded $\mathcal{A}$ -approximation property, then $\mathcal{L}(E, F)$ contains a non-norm-attaining operator.

Proof. Assume that every operator $E \rightarrow F$ attains its norm. If $B_{\mathcal{A}(E, F)}$ were not SOT closed, Lemma 1 would imply that it is not SOT sequentially closed. The unit ball $B_{\mathcal{A}(E, F)}$ is norm closed and convex. By [1, Lemma 3.4], any norm-closed convex subset of $\mathcal{L}(E, F)$ that is not SOT sequentially closed gives the James property for the pair (E, F) . By [1, Theorem A], the James property yields a non-norm-attaining operator $E \rightarrow F$, a contradiction. Therefore $B_{\mathcal{A}(E, F)}$ is SOT closed.

Now assume that (E, F) has the bounded $\mathcal{A}$ -approximation property. Choose $\lambda \geq 1$ such that

$$\mathcal{L}(E, F) \subseteq \lambda \overline{B_{\mathcal{A}(E, F)}}^{\tau_c}.$$

Since the topology of compact convergence is stronger than the SOT,

$$\overline{B_{\mathcal{A}(E, F)}}^{\tau_c} \subseteq \overline{B_{\mathcal{A}(E, F)}}^{\text{SOT}} = B_{\mathcal{A}(E, F)}.$$

Thus every $T \in \mathcal{L}(E, F)$ belongs to $\lambda B_{\mathcal{A}(E, F)} \subseteq \mathcal{A}(E, F)$. Hence $\mathcal{L}(E, F) = \mathcal{A}(E, F)$. The final equivalent formulation is the contrapositive. $\square$

5. VERIFICATION AND SCOPE

The proof is analytic and uses no computation. The key verifier checks are:

- the ideal $\mathcal{A}$ must be norm closed and separably determined in the domain variable;
- the domain E is reflexive so that the separable complementation argument used in [1, Lemma 3.5] applies;
- the approximation hypothesis is bounded and is stated for the pair (E, F) .

This does not solve the full Ostrovskii problem. It says that any positive example must avoid bounded $\mathcal{A}$ -approximation for every proper domain-separably-determined closed ideal $\mathcal{A}$ in the pair-level sense above. For $\mathcal{A} = \mathcal{K}$, the theorem recovers the compact-operator obstruction from the source paper.

6. NOVELTY CHECK

Cheap run indexes had no existing packet for arXiv:2102.06452. Local searches for “separably determined”, “bounded $\mathcal{A}$ -approximation property”, “operator ideal” with “norm attaining”, and “James property” found the source paper and no existing run result with this theorem. Web searches for exact phrases around separably determined operator ideals and non-norm-attaining operators found no separate later answer.

The arXiv source file of [1], in comments not rendered in the published PDF, mentions that separably determined ideals should allow a similar SOT-closure argument. For that reason this packet should be read as a precise reviewable extension of the source method, with modest novelty, rather than a settlement of the original open problem.

REFERENCES

- [1] S. Dantas, M. Jung, and G. Martinez-Cervantes, *On the existence of non-norm-attaining operators*, arXiv:2102.06452.